

PopNavShift: Stress-Testing Social Navigation under Behavioral Population Shift

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Abstract—Social navigation in shared spaces depends in part on how pedestrians respond to robots. Pedestrian behavior varies across individuals and social contexts in clearance preference, yielding, gap acceptance, and walking speed, while navigation algorithms are often evaluated under a fixed distribution of pedestrian behavior. We introduce PopNavShift, a matched simulation framework for stress-testing social navigation strategies under behavioral population shift. We sample 600 synthetic records from MatrAIX Persona 1M, use Gemini 3.7 Flash to obtain standardized responses to robot encounter probes, and deterministically map these responses to bounded pedestrian motion profiles. Identical physical episodes are replayed across eight population conditions and three robot controllers: reactive avoidance, early yielding, and reciprocal collision avoidance. In a matched intervention on the same 202 personas, changing time pressure reverses 22.4% of mean pedestrian-delay orderings and 23.9% of worst-decile delay orderings, compared with 8.6% of robot travel-time orderings. Across all population pairs, pedestrian-burden rankings remain substantially more sensitive than robot travel-time rankings, with larger effects in spatially constrained settings. A second pedestrian dynamics model shows the same qualitative asymmetry. These results show that conclusions about social navigation performance can depend on the pedestrian population represented during evaluation.

I. INTRODUCTION

Social navigation algorithms are evaluated against assumptions about how pedestrians respond to robots in shared space. These responses vary across people and situations. Pedestrians differ in the clearance they maintain, when they yield, how readily they accept small gaps, and how quickly they move around an approaching robot. Human–robot proxemics studies have shown that these behaviors are affected by robot motion, orientation, gaze, and interaction context, while social navigation research treats proxemics, passing conventions, and preferred paths as part of socially appropriate navigation [1]–[4]. In evaluation through simulation, this variation is represented through a particular distribution of pedestrian responses.

Existing methods incorporate pedestrian emotion, individual preferences, and social judgments into planning [5]–[7], while SOCIALGYM and recent pedestrian models expose configurable or heterogeneous behaviors [8], [9]. These tools make it possible to evaluate navigation methods with diverse pedestrians, but comparisons are still usually made within one selected behavioral distribution. This matters because different controllers can place different amounts of adjustment on the robot and nearby pedestrians. A controller that performs well when pedestrians readily yield or accept close passing may compare differently with another controller when those behaviors become less common. The resulting benchmark

conclusion may therefore depend on the represented population even when geometry, density, and initial conditions remain unchanged. Current evaluation guidelines emphasize human, environmental, and task context [10], yet this source of evaluation sensitivity has received limited direct study.

We introduce **PopNavShift**, a matched framework that holds physical episodes and controllers fixed while changing the represented pedestrian population (Fig. 1). Persona 1M records are probed with standardized robot encounters and deterministically mapped to bounded motion profiles. Identical episodes are replayed across population conditions under reactive geometric, early yielding, and Robot-ORCA controllers. A matched intervention additionally changes time pressure for the same persona records. We assess stability through pairwise reversals in controller rankings for robot travel time, mean pedestrian added delay, and worst-decile pedestrian added delay.

This work makes three contributions. First, we treat behavioral population shift as an explicit dimension of social navigation evaluation. Second, we measure population dependence through matched changes in controller rankings and pedestrian burden relative to baselines without a robot. Third, we test whether this dependence persists across population conditions, physical constraints, mappings from responses to motion, and pedestrian dynamics models.

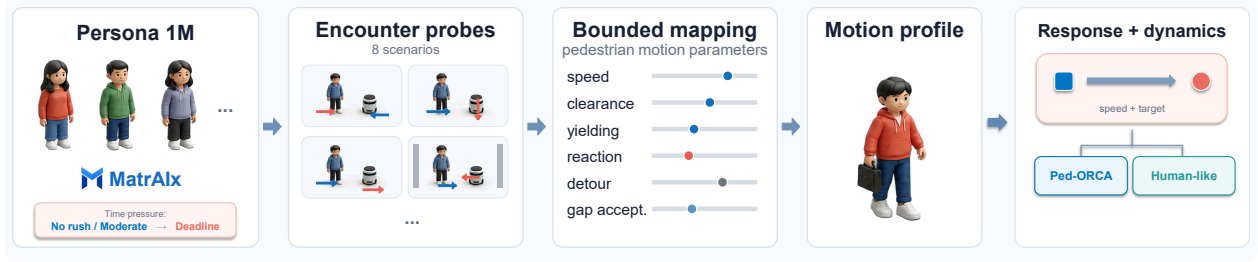
II. RELATED WORK

A. Social Norms and Variation in Human–Robot Proxemics

Social robot navigation depends on how people use and interpret shared space around a robot. Prior work has examined proxemics, perceived safety, passing conventions, preferred paths, and socially acceptable robot behavior as components of navigation in human environments [4], [10]. Human–robot proxemics studies further show that the spacing people maintain around robots varies with properties of the encounter. Robot approach direction, orientation, gaze, and motion can affect the distance people adopt during interaction [1]–[3]. These findings indicate that pedestrian spatial behavior depends on both encounter geometry and interaction cues.

Social conventions can also vary across populations. Eresha et al. examined cultural differences in proxemic behavior around humanoid robots [11], while Xiao et al. compared pedestrian collision-avoidance behavior across participants in the United States and Israel [12]. Such variation matters for navigation because preferred passing distance, yielding, and acceptance of close interaction affect how space is shared between a robot and nearby pedestrians. Recent

a From persona records to pedestrian motion



b Matched replay of one encounter

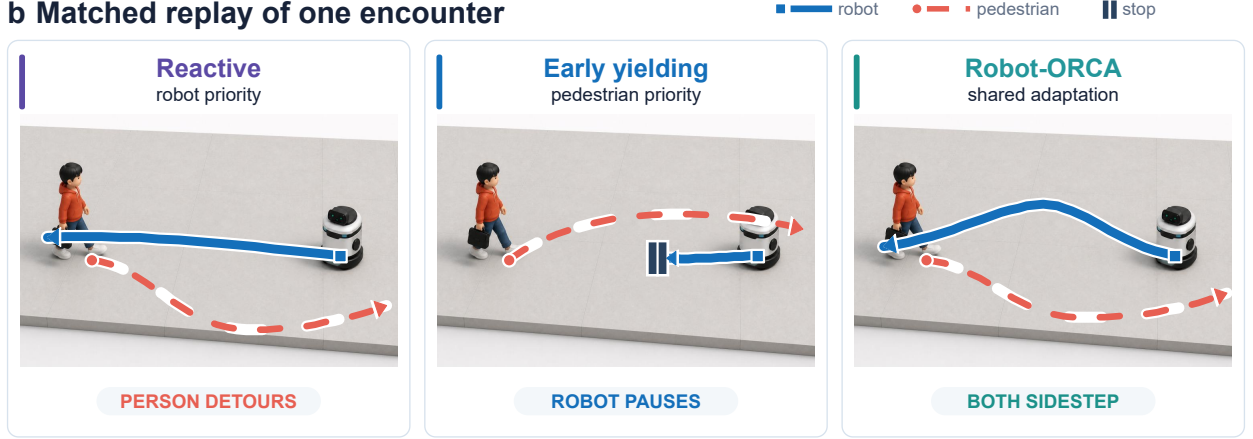


Fig. 1. PopNavShift evaluates social navigation strategies under behavioral population shift. (a) Persona records are converted through standardized robot encounter probes and a deterministic bounded mapping into pedestrian motion profiles. The matched intervention changes time pressure for the same persona records before repeating the probe-to-motion pipeline. (b) Identical physical encounters are replayed under reactive geometric, early yielding, and Robot-ORCA controllers. Ped-ORCA is the primary pedestrian dynamics backend; navground Human-like behavior is used for robustness analysis.

social navigation guidelines likewise emphasize that socially appropriate behavior depends on the human, environmental, and task context in which navigation occurs [10].

Other work incorporates human variation directly into robot navigation. EWareNet adapts intent prediction and spatial profiles using inferred pedestrian emotion [5]; Choi et al. infer user-specific navigation preferences [6]; and NaviDIFF incorporates human preferences and social principles into navigation behavior [7]. These approaches primarily study how a robot can adapt to a person or inferred preference. PopNavShift instead examines whether conclusions about the relative performance of fixed robot controllers remain stable when the represented distribution of pedestrian behavior changes.

B. Pedestrian Diversity in Social Navigation Modeling

Pedestrian motion models provide the human side of simulated social navigation encounters. Classical approaches such as the Social Force model and reciprocal collision avoidance generate pedestrian motion from explicit interaction rules [13], [14]. Other approaches learn crowd motion or interaction behavior from observed trajectories. In many simulation settings, however, pedestrians still share a common motion model or vary within a predefined parameterization.

Recent systems make behavioral diversity more explicit. SOCIALGYM exposes configurable human parameters for

social navigation experiments [8]. SEAN 2.0 supports varied social situations through multiple methods for specifying and generating pedestrian motion [15]. MAC-ID models individual differences through a local coordination factor that changes pedestrian behavior from more competitive to more cooperative interaction, and its BSON benchmark uses these behaviors to construct diverse social environments for navigation evaluation [9]. These methods demonstrate that pedestrian behavior can be varied systematically within simulation and provide tools for testing navigation methods beyond a single pedestrian type.

Recorded datasets provide a complementary source of observed human motion. SCAND contains human demonstrations of socially compliant robot navigation [16], while THÖR and THÖR-MAGNI provide accurately tracked human motion and robot encounters [17], [18]. These datasets are useful for characterizing observed motion and setting plausible physical ranges; in our experiments, reported THÖR statistics inform walking-speed, spacing, and encounter ranges. They do not provide the structured persona attributes and field-level interventions required for the matched population shifts considered here.

C. Evaluation under Behavioral Population Shift

Social navigation benchmarks standardize comparisons across environments and interaction conditions. SocNavBench

replays recorded pedestrian trajectories under a common protocol [19], SocNav1 provides socially judged navigation situations [20], and SOCIALGYM varies human and environmental parameters systematically [8]. Evaluation guidelines likewise emphasize testing navigation methods across relevant human, environmental, and task contexts [10].

BSON is especially close to our setting because it evaluates navigation methods in environments with diverse pedestrian interaction styles [9]. PopNavShift instead treats the behavioral population distribution as the comparison variable itself. Physical episodes and controller definitions remain matched while the pedestrian-profile distribution changes, and population dependence is measured through changes in controller ordering across those distributions.

III. METHOD

A. Framework Overview

PopNavShift separates the construction of behavioral populations from pedestrian dynamics and robot control (Fig. 1). Eight fixed robot encounter probes elicit structured responses from each persona, which a deterministic mapping converts into seven bounded motion controls. Eight population conditions define distributions over these profiles for matched encounters using a shared pedestrian response layer and a continuous dynamics model.

B. Persona Records and Standardized Probes

We sample 600 records from the public MatRAIx Persona 1M release [21], [22]. The prompt includes available fields related to time pressure, safety and risk preferences, decision style, trust in technology, patience, accessibility needs, and mobility. Demographic attributes are excluded from the prompt and mapping.

Gemini 3.7 Flash receives each record and answers eight fixed robot encounter probes. For each encounter, it selects one action from *continue*, *slow*, *stop*, *yield*, *pass*, or *detour*. The probes vary relative approach direction, available space, crowding or obstruction, and robot behavior. Together, they cover situations in which clearance, yielding, speed adjustment, and detouring can affect the interaction.

After selecting the encounter actions, the model returns an ordered response block covering preferred clearance, willingness to wait, yielding, small-gap acceptance, detouring, robot trust, and urgency on five-level scales. The complete structured response is cached for each persona and reused across simulation scenarios. The probes form a response-elicitation battery; they are separate from the physical scenarios used in the simulation experiments.

C. Mapping from Responses to Motion

The probe responses are converted into continuous pedestrian controls using a deterministic bounded mapping. The five ordered response levels are encoded as $\{0, 0.25, 0.5, 0.75, 1\}$. For persona i , the motion profile is

$$\theta_i = (v_i, d_i, a_i, y_i, \tau_i, L_i, g_i),$$

TABLE I
BEHAVIORAL POPULATION CONDITIONS AND COHORT SIZES

Condition	Construction	n
Homogeneous	Overall median parameter vector	1
Mixed	All mapped records	600
High time pressure	Deadline or Emergency source field	49
Mobility support	Explicit mobility or motor-support field	45
High caution	Upper quartile of clearance/yield/gap score	170
Low robot trust	Low or very-low probe response	197
Matched base	Original no-rush/moderate records	202
Matched deadline	Same IDs; time pressure set to Deadline	202

where the components denote preferred speed, robot clearance, avoidance strength, yielding propensity, reaction delay, detour threshold, and gap acceptance, respectively. These are operational simulator controls rather than estimated psychological traits.

Let q_i^c , q_i^w , q_i^y , q_i^g , q_i^d , q_i^t , and q_i^u denote the normalized responses for clearance, waiting, yielding, small-gap acceptance, detouring, robot trust, and urgency. Mapping A also summarizes the categorical encounter actions using fixed ordered scores, whose mean is q_i^a . Its caution score is

$$C_i = 0.34q_i^c + 0.22(1 - q_i^g) + 0.22q_i^y + 0.22q_i^a. \quad (1)$$

The normalized controls combine these responses according to their intended simulation roles: urgency affects preferred speed and yielding; clearance and gap acceptance use their corresponding responses directly; avoidance combines caution with robot trust; reaction delay combines urgency and trust; and detour tendency combines the reported detour response with caution. All coefficients are fixed before simulation and are not fitted to observed trajectories.

Each normalized control z_i^k is clipped to $[0, 1]$ and mapped to its physical range $[\ell_k, h_k]$ by

$$\theta_i^k = \ell_k + z_i^k(h_k - \ell_k). \quad (2)$$

The physical ranges are 0.60–1.80 m/s for preferred speed, 0.30–1.20 m for robot clearance, 0.60–2.20 for avoidance strength, 0.15–1.20 s for reaction delay, and 1.0–4.0 m for the detour threshold. Yielding and gap acceptance remain in $[0, 1]$. Mobility-related fields apply ordered speed caps after mapping. The resulting speed and spacing ranges overlap values reported in THÖR and related datasets [17].

We additionally test three alternative mappings from responses to motion. Mappings B and C modify the weighting and action-response construction, while Mapping D uses separable response dimensions without weighted mixtures. All four mappings retain the same physical ranges and mobility caps. Mapping A is used in the main experiment and Mappings B–D are used for sensitivity analysis.

D. Population Conditions and Matched Intervention

Table I defines the eight population conditions used in the experiments. These selected behavioral contrasts can overlap in persona membership. Figure 2(a) summarizes the resulting distributions over the motion parameters.

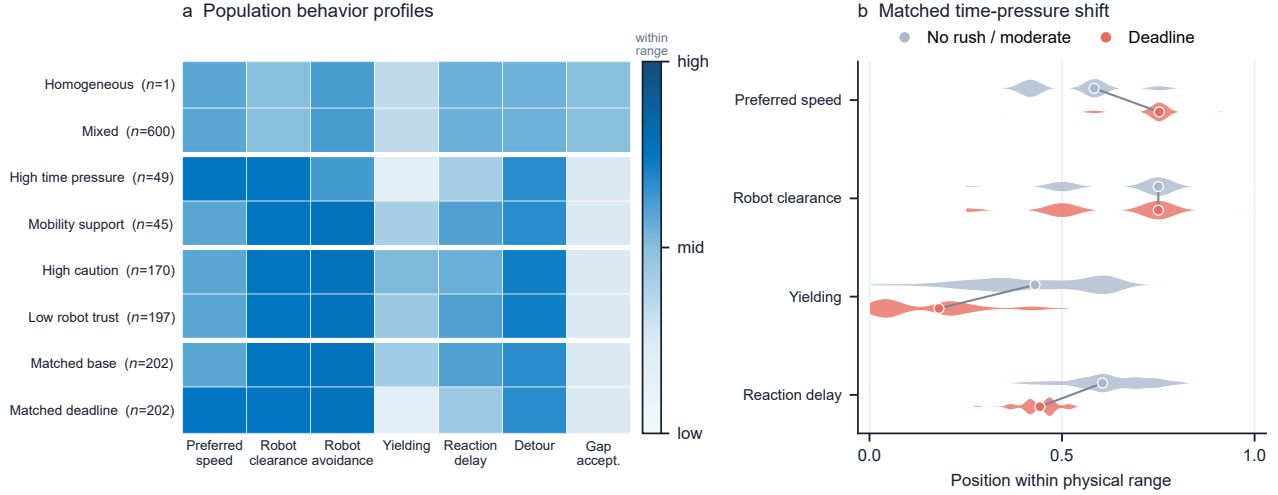


Fig. 2. Pedestrian motion profiles across population conditions. (a) Median position within each parameter’s physical range; labels report cohort size. White separators distinguish reference conditions, source-field cohorts, probe-defined cohorts, and the matched pair. (b) Profile distributions for the same personas before and after the matched change in time pressure from *No rush/Moderate* to *Deadline*.

The matched time-pressure intervention provides the main controlled population shift. It contains the same 202 persona records in both conditions: 113 originally have *No rush* and 89 have *Moderate* time pressure. The matched-base condition uses the original records. For the matched-deadline condition, only the time-pressure field is changed to *Deadline*, after which the same encounter probes and mapping from responses to motion are rerun. Persona identity and all other source fields remain unchanged.

Time pressure is used as a mutable contextual variable because controlled pedestrian studies have found that imposed time pressure can change crossing speed and gap acceptance [23]. This evidence motivates the intervention; the observed effect sizes are not used to calibrate the persona responses or motion parameters. Figure 2(b) shows how the resulting motion-profile distributions change for the same personas.

For episodes containing multiple pedestrians, profiles are sampled with replacement using fixed episode-specific assignments. Matched-base and matched-deadline runs use the same persona at every pedestrian slot. Any difference between the two conditions therefore originates from responses changed by the time-pressure intervention rather than from different sampled identities.

E. Pedestrian Response and Dynamics

Each motion profile enters simulation through a shared robot-conditioned pedestrian response layer. The response activates after the robot enters sensing range and the profile-specific reaction delay has elapsed. Both agents are projected over a four-second horizon using the effective clearance

$$d_i^{\text{eff}} = d_i(1 - 0.35g_i).$$

A conflict is predicted when the closest projected center distance falls below the sum of the two agent radii and d_i^{eff} .

During a predicted conflict, yielding propensity reduces the pedestrian’s preferred speed as the robot approaches the

detour threshold L_i . When the robot is nearly stationary, this reduction is relaxed to allow the pedestrian to pass. Inside L_i , the response layer places a lateral waypoint 1.5 m ahead, with offset determined by effective clearance and avoidance strength. Each episode fixes a passing side and switches sides if an obstacle blocks the preferred route.

Navground provides the continuous two-dimensional simulation [24]. The primary pedestrian backend, Ped-ORCA, uses omnidirectional Optimal Reciprocal Collision Avoidance (ORCA) with a 4.5 m spatial horizon [14]. A robustness experiment uses navground’s Human-like (HL) behavior, which responds to visual collision cues within a 5 m horizon [25]. Both backends receive the same preferred speeds and targets from the shared response layer. The simulation step is 0.1 s and the episode time limit is 45 s.

F. Robot Controllers

The robot is modeled as a differential-drive disk with radius 0.35 m. All controllers share the same goal, footprint, dynamic limits, local observations, and collision geometry.

The *reactive geometric* controller samples feasible wheel commands over a 2 s horizon with a 0.04 m safety buffer and adjusts its trajectory when a conflict becomes imminent. The *early yielding* controller detects conflicts over a 4.5 s horizon, moves toward a clear pull-aside pose with a 0.16 m buffer, and waits while the pedestrian crosses the conflict region. The *Robot-ORCA* controller uses reciprocal collision avoidance with a 5 m spatial horizon and a 0.12 m safety margin under navground’s two-wheel kinematics.

The controllers therefore differ primarily in how interaction adjustment is allocated. Reactive control postpones robot-side adjustment, early yielding gives greater priority to the pedestrian, and Robot-ORCA uses reciprocal adaptation. Ped-ORCA denotes the pedestrian dynamics backend; Robot-ORCA denotes the reciprocal robot controller.

IV. EXPERIMENTAL DESIGN

A. Matched Evaluation Grid

We evaluate PopNavShift on five types of physical encounters: a bidirectional sidewalk, a head-on encounter, a crossing, a bottleneck, and a static obstruction. The scenario grid varies available width (1.8, 2.4, 3.0, and 4.2 m), pedestrian density (0.06, 0.12, and 0.20 persons/m²), and robot speed (0.60 and 1.00 m/s). Head-on encounters contain one pedestrian and use the lowest density, yielding 104 valid physical cells rather than a full Cartesian product.

Each physical cell contains three fixed variants with different pedestrian start positions and walking directions, giving 312 episodes in total. Every episode is replayed with all eight population conditions and all three robot controllers. For each population–episode combination, an additional run without a robot provides the pedestrian baseline. The matched design keeps physical geometry, episode initial conditions, and controller definitions unchanged when the represented pedestrian population changes.

B. Metrics and Ranking Stability

Robot-centered outcomes include success, travel time, path length, stopping time, pedestrian and world collisions, and minimum pedestrian clearance. Pedestrian-centered outcomes include travel time, walking distance, stopping time, detour distance, and minimum robot clearance.

Our main pedestrian-burden measure is added travel time. For pedestrian j ,

$$\Delta T_j = T_j^{\text{robot}} - T_j^{\text{no robot}},$$

where the run without a robot uses the same physical episode and pedestrian assignment. Added distance is defined analogously. A paired delay is available when the pedestrian completes both runs.

A physical cell fixes scenario type, width, density, and robot speed and pools its three episode variants. Mean pedestrian delay is computed over all available paired records in the cell. To capture concentrated burden, worst-decile delay is the mean of the largest $\lceil 0.1n \rceil$ paired delays. Failed or timed-out robot runs retain the 45 s episode limit as robot travel time.

For each physical cell and metric, we compare every pair of controllers separately under each population. A *ranking reversal* occurs when a non-tied controller ordering under one population changes sign under another. We use tie tolerances of 0.02 s for pedestrian delay and 0.05 s for robot travel time. We additionally report a completed-cell analysis restricted to cells in which all three controllers finish every episode variant under both populations being compared.

To quantify the magnitude of population effects independently of controller ordering, we define *population sensitivity* as the range of a metric across population conditions for a fixed controller and physical cell.

C. Sensitivity Analyses

We test whether the observed population dependence is specific to the mapping from responses to motion or pedestrian

dynamics model. Mapping sensitivity is evaluated on twelve representative episodes spanning narrow and wide sidewalks, bottlenecks, and static obstructions. Mappings A–D use the same cached responses, population membership, and episode assignments. In addition to ranking-reversal rates, we measure agreement with Mapping A on eligible non-tied controller comparisons.

We also perturb the structure of the pedestrian profiles. One analysis reassigns complete motion-profile vectors across persona identities before reconstructing five shared populations. A second independently shuffles each motion-parameter column within a population, preserving marginal parameter distributions while breaking their cross-parameter association. Separate runs isolate the contributions of time-pressure, caution, and trust response blocks.

Finally, we repeat a subset of the evaluation with nav-ground’s Human-like pedestrian dynamics. This comparison uses twelve obstacle-free sidewalk, head-on, and crossing episodes and four shared populations: homogeneous, mixed, matched base, and matched deadline. Physical scenes, persona assignments, response rules, and robot controllers remain matched between Ped-ORCA and Human-like dynamics. Backend comparisons use the intersection of cells completed by all three controllers under both models, after which the same metric-specific tie tolerances are applied separately to each backend.

V. RESULTS

A. Matched Time-Pressure Shift

The main experiment contains 7,488 robot runs and 2,496 matched runs without a robot, with paired pedestrian delay available for nearly all records. Changing time pressure from *No rush/Moderate* to *Deadline* alters at least one categorical response for 201 of the 202 matched personae.

This shift changes controller comparisons more often for pedestrian burden than for robot travel time. Mean-delay orderings reverse in 22.4% of comparable cells and worst-decile-delay orderings in 23.9%, compared with 8.6% for robot travel time. The effect also differs across controllers. As shown in Fig. 3, the matched deadline shift reduces pedestrian delay under reactive control and Robot-ORCA but increases it under early yielding. The population change therefore alters the relative burden associated with different navigation strategies rather than producing a common shift across controllers.

B. Comparisons across Population Pairs

The same pattern appears across all 28 population pairs. Table II pools the three controller pairs after excluding ties. Robot travel-time orderings reverse in 8.7% of eligible comparisons, compared with 25.8% for mean pedestrian delay and 24.8% for worst-decile delay.

Restricting the analysis to cells in which all three controllers complete every episode under both populations reduces the reversal rates to 5.0%, 21.5%, and 22.1%, respectively. The higher sensitivity of pedestrian-burden rankings

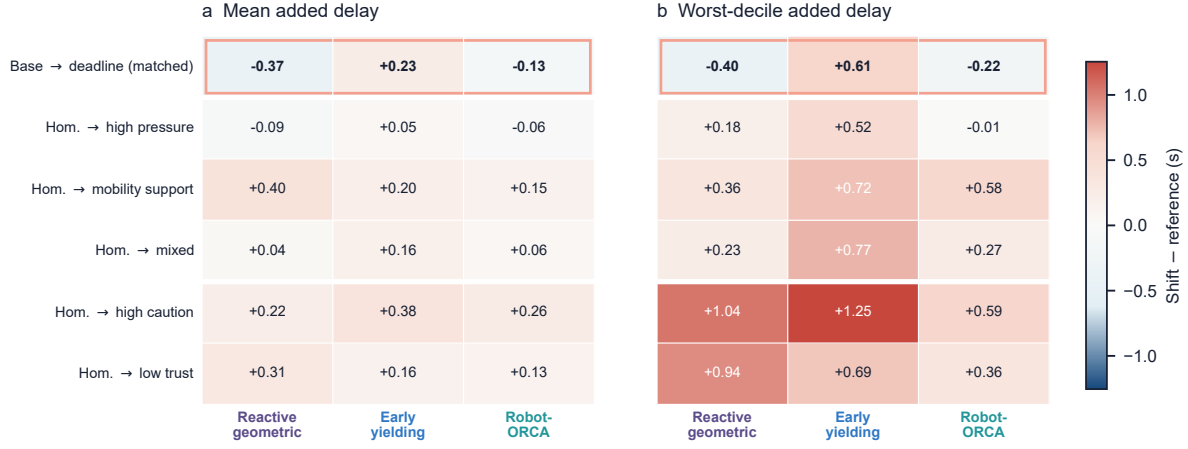


Fig. 3. Changes in pedestrian delay under behavioral population shift. Entries show shifted minus reference delay over the same 104 physical cells. The outlined row shows the matched time-pressure intervention on the same personas. White separators distinguish the matched intervention, source-field cohorts, the mixed population, and probe-defined cohorts.

therefore remains after controller failures are removed from the comparison.

C. Aggregate Controller Outcomes

The controllers also exhibit distinct aggregate performance trade-offs. Table III summarizes outcomes over the full matched grid. Reactive control achieves the highest completion rate at 90.3%, while Robot-ORCA produces the lowest mean pedestrian delay (0.65 s) and worst-decile delay (2.80 s). Early yielding has the lowest mean delay among the two non-reciprocal controllers but also the highest timeout rate.

Robot-ORCA performs best on aggregate pedestrian delay, whereas reactive control completes more runs. Most Robot-ORCA collisions are contacts with the environment rather than pedestrians, accounting for the difference between its pedestrian- and world-collision rates in Table III.

D. Spatial Constraint and Tail Burden

Pedestrian burden is concentrated in the tail. Across the controllers, worst-decile pedestrian delay is approximately four times the corresponding population mean. Population shifts can also affect the tail differently from the mean, as illustrated by the early yielding response to the matched time-pressure intervention in Fig. 3.

Available space strongly affects the magnitude of population sensitivity. As shown in Fig. 4, the range of mean pedestrian delay across populations is larger on narrow sidewalks and bottlenecks. Pooled population sensitivity decreases from 2.23 s at 1.8 m width to 0.50 s at 4.2 m, with

TABLE II
REVERSALS IN CONTROLLER RANKINGS ACROSS POPULATION PAIRS

Metric	All cells	Completed cells
Robot travel time	8.7%	5.0%
Mean pedestrian delay	25.8%	21.5%
Worst-decile delay	24.8%	22.1%

TABLE III
CONTROLLER OUTCOMES OVER THE FULL MATCHED GRID

Outcome	Early		
	Reactive	yielding	Robot-ORCA
Success (%)	90.3	81.5	87.8
Pedestrian collision (%)	2.6	5.4	2.1
World collision (%)	0.1	0.0	7.1
Timeout (%)	7.0	13.0	3.1
Robot travel time (s)	22.9	28.4	24.2
Minimum clearance (m)	0.70	0.70	0.73
Mean added delay (s)	1.25	1.06	0.65
Worst-decile delay (s)	5.13	4.42	2.80

TABLE IV
RANKING REVERSALS AND AGREEMENT ACROSS MAPPINGS FROM RESPONSES TO MOTION

Mapping	Reversal rate		Agreement with A	
	Mean	Tail	Mean	Tail
A: original	24.5%	26.0%	—	—
B: ordered	24.2%	28.6%	92.3%	92.6%
C: equal weights	24.6%	25.7%	92.2%	91.8%
D: direct	37.5%	35.8%	76.8%	71.0%

the same overall direction for each controller. When analysis is restricted to cells completed by all controllers, width is no longer monotonic, indicating that part of the stronger narrow-space effect coincides with controller failures.

E. Mapping and Profile Structure

The population effect persists when the mapping from responses to motion is changed. Table IV compares the four mappings on the representative subset. Mappings B and C preserve more than 91% of Mapping A's mean- and tail-delay controller orderings and produce similar reversal rates. The simpler direct Mapping D yields higher reversal rates, reaching 37.5% for mean delay and 35.8% for worst-decile delay.

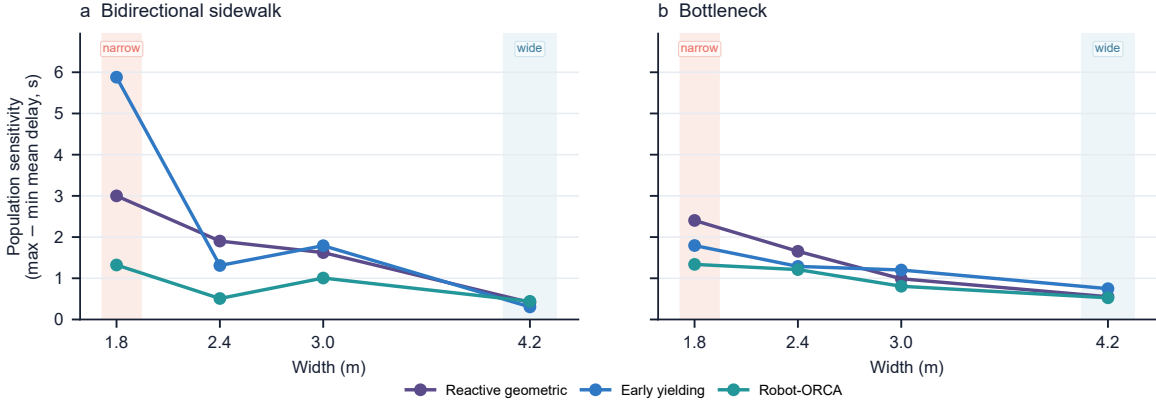


Fig. 4. Population sensitivity across available widths. Population sensitivity is the range of mean pedestrian delay across population conditions for a fixed controller and physical cell, shown for (a) bidirectional sidewalks and (b) bottlenecks.

Changing the association structure among persona profiles also changes the magnitude of the effect. Across five shared populations, randomly reassigning complete motion profiles among persona identities reduces mean/tail reversal rates from 28.4%/26.9% to 19.8%/18.8%. Independently shuffling each parameter column gives 26.8%/29.7%. Thus, shifted parameter distributions continue to produce ranking changes after profile associations are disrupted, while the associations themselves affect how frequently reversals occur. In the single-block analyses, time pressure and caution produce more ordering changes than robot trust alone.

F. Pedestrian-Dynamics and LLM Robustness

The qualitative difference between pedestrian burden and robot travel time also appears under a second pedestrian dynamics model. On cells completed by all three controllers under both backends, Ped-ORCA produces mean- and worst-decile-delay reversal rates of 10.5% and 9.8%, compared with 3.2% for robot travel time. Under the Human-like backend, the corresponding rates are 35.2%, 34.6%, and 2.4%. The two dynamics models differ substantially in reversal magnitude, while both produce more pedestrian-burden ranking reversals than robot travel-time reversals.

Repeating the matched intervention with Gemini 3.5 Flash Lite gives the same qualitative pattern. With the same persona IDs, reversal rates are 25.6% for mean pedestrian delay, 26.5% for worst-decile delay, and 9.1% for robot travel time. Probe-defined cohort membership can change across language models, so this comparison is restricted to the matched intervention.

VI. DISCUSSION

A. Population-Conditioned Evaluation

The results show that social navigation controller comparisons can depend on the behavioral population represented during evaluation. With the physical episode and controller definitions held fixed, changes in pedestrian motion profiles frequently alter pedestrian-burden rankings. This extends prior work on human variation in social navigation by showing

that behavioral diversity can affect the relative evaluation of controllers, not only the simulated interaction itself.

Pedestrian burden is more population-sensitive than robot travel time because shifts in speed, clearance, yielding, reaction, detouring, and gap acceptance interact differently with controller behavior. The population-specific results suggest corresponding design priorities: faster establishment of clear passage under time pressure, less forced slowing or detouring for mobility-support pedestrians, and larger usable clearance and earlier, more predictable conflict resolution for cautious or low-trust pedestrians. These differences become more consequential in narrow sidewalks and bottlenecks.

B. Implications for Social Navigation Evaluation

Behavioral population composition should therefore be treated as an evaluation dimension alongside geometry, density, and other scenario conditions. A benchmark can contain heterogeneous pedestrians while still evaluating every controller under one fixed mixture of behaviors; this may hide population-specific changes in relative performance. Matched population replay provides a direct way to test this dependence by evaluating the same physical episodes under alternative pedestrian distributions and reporting robot-centered outcomes together with mean and tail pedestrian burden.

This framework can also be used with empirically measured populations. Future datasets that combine contextual attributes with observed human-robot motion could provide grounded distributions of clearance, yielding, speed, and gap acceptance for different deployment settings, allowing population robustness to be evaluated with observed rather than synthetic behavioral variation.

VII. LIMITATIONS

The behavioral populations in this study are synthetic and should be interpreted as controlled simulation inputs rather than estimates of particular cities, cultures, or demographic groups. The persona-to-motion pipeline is not calibrated against joint observations of persona attributes, encounter responses, and pedestrian motion from the same participants.

The matched time-pressure intervention is likewise a controlled perturbation of the synthetic records and should not be interpreted as an estimate of the causal effect of deadlines on real pedestrians. Alternative mappings, pedestrian dynamics, and language models preserve the main qualitative pattern while changing reversal magnitudes and some controller orderings.

The simulation also abstracts several aspects of real pedestrian–robot interaction. Pedestrians are represented as disk agents, assistive-device geometry and group behavior are not modeled, and the robot receives perfect local state. The Human-like dynamics analysis is limited to a smaller obstacle-free subset. Future work can combine empirically measured behavioral populations with richer pedestrian models and real-world robot experiments.

VIII. CONCLUSION

We introduced PopNavShift to test whether social navigation controller comparisons remain stable when the represented pedestrian population changes. Across matched physical episodes, pedestrian burden rankings were more sensitive to behavioral population shift than rankings based on robot travel time, and population sensitivity was greater in spatially constrained settings. Changing only time pressure for the same persona records frequently reversed controller orderings for mean and worst-decile pedestrian delay. The same qualitative pattern persisted across alternative mappings, pedestrian dynamics, and language models. These results suggest that behavioral population composition should be treated as an explicit dimension of social navigation evaluation when comparing navigation strategies.

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